

Vidush Singhal

/vi.dʊʃ 'siŋ.gʱəl/

West Lafayette, IN, USA (Open to full time roles)

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Summary

Ph.D. student in Electrical and Computer Engineering at Purdue University specializing in compiler infrastructure, program analysis, and performance optimization. Experienced with LLVM/Clang, MLIR/Torch-MLIR, GPU compiler tooling, automatic parallelization, data layout optimization, and software verification.

Experience

Microsoft Research (RiSE)

Research Intern

Redmond, WA

May 2025 – Aug 2025

- Built **C2Pulse**, an LLVM/Clang-based transpiler for verifying memory correctness of annotated C programs.
- Implemented a Clang-to-Clang source transformer (**~7K LOC of C++**) that emits Pulse code for F* verification.
- Built a C benchmark suite and verified C components of DPE (Dice Protection Environment) through C2Pulse.

Lawrence Livermore National Laboratory (LLNL)

Computing Scholar Intern

Livermore, CA

May 2024 – Aug 2024

- Engineered an LLVM Attributor pass to optimize byte layout within stack allocations for improved spatial locality.
- Implemented LLVM IR analyses and transformations to eliminate dead bytes and reorder frequently accessed bytes.
- Reduced XSBench runtime by **6%** when running under the sanitizer.

Nod.ai Labs (Acquired by AMD)

Machine Learning Compiler Engineering Intern

Santa Clara, CA

May 2022 – Aug 2022

- Implemented Torch-MLIR kernels and compiler support for multiple PyTorch operators.
- Improved the PyTorch-to-MLIR compilation flow by expanding operator coverage.
- Added support for the Facebook DLRM recommendation model in the PyTorch-to-MLIR pipeline.
- Investigated ILP solver improvements for the Alpa scheduling framework.

Purdue University - PLCL Group

Graduate Research Assistant

West Lafayette, IN

May 2021 – Present

- **Gibbon Compiler**
 - Optimized layouts of recursive datatypes to improve spatial locality.
 - Measured **speedups of 1.14x to 54x** over the best prior work.
 - Implemented an array-of-structs to structure-of-arrays-style transformation for packed algebraic datatypes.
 - Enabled selective traversal with **1.1x to 12x speedups** and ongoing loopification/vectorization optimizations.
- **Orchard Compiler**
 - Developed a compiler framework to automatically fuse and parallelize tree traversals.
 - Measured **1x to 5x speedups** over baseline tree traversals.
- **Cornucopia**
 - Developed a fuzzing-based framework to generate optimized binary corpora using `llc` optimization flags.
 - **Discovered bugs** in LLVM compiler optimization passes.
 - **Uncovered decompilation failures** in angr, Ghidra, and related binary analysis tools.
 - Embedded ground-truth metadata in generated binaries to evaluate binary analysis tools.
- **Copse Compiler**
 - Developed a framework for vectorized evaluation of fully homomorphic encryption (FHE) decision trees.

Senior Design Project - SoCET Team

Undergraduate Research

West Lafayette, IN

Jul 2020 – May 2021

- Developed an LLVM-based compiler tool for proprietary RISC-V hardware.
- Implemented multiple LLVM RISC-V backend Machine IR passes to identify sparse program regions.
- Annotated sparse regions so hardware could detect them at runtime and perform runtime checks.

Education

Purdue University, West Lafayette, IN

Ph.D. in Electrical & Computer Engineering

Expected December 2026

M.S. in Electrical & Computer Engineering

May 2023

B.S. in Computer Engineering

May 2021

Relevant Coursework: Compiler Code Generation and Optimization, Compilers for GPUs, Programming Parallel Machines, Reasoning About Programs, Programming Languages, Formal Languages/Computability/Parallelization, Holistic Software Security

Publications

- Raghav Malik, **Vidush Singhal**, Benjamin Gottfried, Milind Kulkarni. “Vectorized Secure Evaluation of Decision Forests.” [PLDI 2021](#).
- **Vidush Singhal**, Akul Abhilash Pillai, Charitha Saumya, Milind Kulkarni, Aravind Machiry. “Cornucopia: A Framework for Feedback Guided Generation of Binaries.” [ASE 2022](#).
- **Vidush Singhal**, Chaitanya Koparkar, Joseph Zullo, Artem Pelenitsyn, Michael Vollmer, Mike Rainey, Milind Kulkarni. “Optimizing Layout of Recursive Datatypes with Marmoset.” [ECOOP 2024](#).
- **Vidush Singhal**, Laith Sakka, Kirshanthan Sundararajah, Ryan Newton, Milind Kulkarni. “Orchard: Heterogeneous Parallelism and Fine-grained Fusion for Complex Tree Traversals.” [ACM TACO 2024](#).
- Chaitanya S. Koparkar, **Vidush Singhal**, Aditya Gupta, Mike Rainey, Michael Vollmer, Artem Pelenitsyn, Sam Tobin-Hochstadt, Milind Kulkarni, Ryan R. Newton. “Garbage Collection for Mostly Serialized Heaps.” [ISMM 2024](#).

Skills

Languages: C, C++, Haskell, Python, Java, Bash, CUDA, Assembly (ARM, x86), SystemVerilog

Compilers & Verification: LLVM, Clang, MLIR/Torch-MLIR, GCC, Coq, Dafny, F*, Pulse

Parallel, Systems & ML: OpenMP, MPI, Pthreads, OpenCilk, Intel TBB, gem5, AFL++, GDB, Linux, PyTorch, TensorFlow, NumPy

Awards, Grants & Service

Dean’s List and Semester Honors, Purdue University

(2017 - 2020)

LLVM Developers’ Meeting Travel Grant

2024

Research Service - AEC/PC/Reviewer for PLDI, CGO, ICFP, PPOPP, POPL, SPLASH, C3PO, JOSS (2023–2026)

Seminar Co-Coordinator, [Purdue Programming Languages and Systems Research Group](#)

(2025 - present)